

EAGLE NS80 Centrifugal Pump coupled to a KUBOTA RK70-S Diesel Engine

Test Report No.	:	2013 - 120
Date of Test	:	September 30, 2013
Test Valid Until	:	September 30, 2018
Brand	:	EAGLE
Model	:	NS80
Serial No.	:	016860
Type and Size	:	Single-end suction, 75 x 75 mm, Non-Self Priming Centrifugal pump
Manufacturer	:	SUPERCAST FOUNDRY & MACHINERY CORP. P. Gregorio St., Lingunan, Valenzuela City
Test Requested by	:	KUBOTA PHILS., INC. 232 Quirino Highway, Baesa, Quezon City 1106 Metro Manila
Machine Classification	:	Commercial

SUMMARY

The EAGLE NS80 centrifugal pump coupled to a KUBOTA RK70-S diesel engine was tested for performance and suction condition. It was tested at the recommended full throttle speed setting of the engine. The engine and pump both used double groove B-section pulleys with actual diameters of 102 mm (4.0 inches) and 88.9 mm (3.5 inches), respectively and two pieces of V-belts. The engine and pump were mounted on a steel base.

The Maximum Total Head (H) was 24.7 m at zero discharge condition. The shaft speed of the pump and engine at this point were 2475 and 2173 rpm, respectively with Engine Fuel Consumption Rate of 2.12 L/h. The Maximum Discharge Capacity of 13.8 L/s occurred at 5.00 m Total Head with Maximum Fuel Consumption Rate of 1.99 L/h. The shaft speed of the pump and engine at this point were 2075 and 1818 rpm respectively. The Maximum System Efficiency of 5.88% occurred at 8.11 L/s Discharge Capacity.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 7.47 m. The Net Positive Suction Head Available at this point was 2.40 m.

PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pump performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. The pump was mounted on the laboratory test set-up shown in **Figure 1**.

The test was conducted on September 30, 2013 at the AMTEC Test Laboratory, UPLB, College, Laguna.



Figure 1. EAGLE NS80 Centrifugal pump coupled to KUBOTA RK70-S Diesel engine on the test set-up

METHOD OF TEST

The pumpset was tested following the “Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test”.

DESCRIPTION OF THE PUMP

The EAGLE NS80 Centrifugal pump coupled to KUBOTA RK70-S Diesel Engine shown in **Figure 2**, was a single-end suction, 75 mm x 75 mm, centrifugal pump coupled to a 5.22 kW (7.0 Hp) KUBOTA RK70-S diesel engine. The pump and engine were mounted on a steel base. The power from the engine was transmitted to the pump shaft through a double groove B-section pulley and V-belts. The actual diameter of the engine and pump pulleys were 102 mm (4.0 inches) and 88.9 mm (3.5 inches), respectively. The suction inlet and discharge outlet were threaded and designed for a 75 mm diameter pipe. The impeller was a closed type made of brass material connected to the pump shaft through a thread. It was housed in a cast iron volute casing. The priming inlet was located on top of the pump beside the discharge outlet. The discharge outlet was located on top of the pump facing upward. The stuffing box used a packing seal. The pump shaft rotated on ball bearing kept in place by a bearing housing integral to the casing.

The prime mover, a 5.22 kW KUBOTA RK70-S diesel engine was a four-stroke, water-cooled, single horizontal cylinder diesel engine. Fuel feed system was of gravity type with the fuel tank located on top of the engine. It utilized an oil-bath type air cleaner. Lubrication system was force-feed lubricating type. Cooling was done by water circulating from cylinder jacket to the radiator using thermo-siphon principle. A fan driven by the pulley on the flywheel was used to cool the warm water in the radiator. It was started using manual cranking lever aided by a manually actuated decompression lever. A muffler was used in the exhaust system. **Table 1** shows the detailed specifications of the pump.

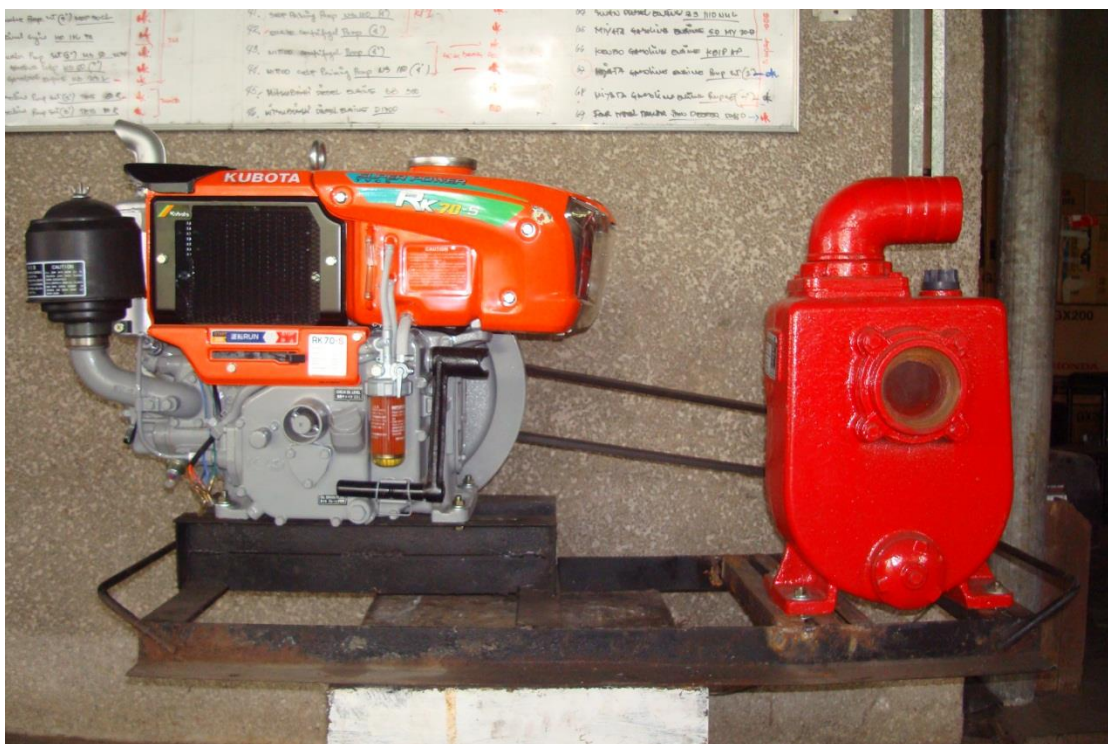


Figure 2. EAGLE NS80 Centrifugal Pump coupled to a KUBOTA RK70-S Diesel Engine

PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the EAGLE NS80 Centrifugal Pump coupled to KUBOTA RK70-S Diesel Engine

1.	Brand	:	EAGLE
2.	Model	:	NS80
3.	Serial No.	:	016860
4.	Make	:	SUPERCASE FOUNDRY & MACHINERY CORP. P. Gregorio St., Lingunan, Valenzuela City
5.	Type	:	Single-end suction, Non-self priming Centrifugal pump
6.	Size (mm)	:	75 x 75
7.	Overall Dimension and Weight (Including mounting frame and engine)		
7.1	Length (mm)	:	1220
7.2	Width (mm)	:	500
7.3	Height (mm)	:	582
7.4	Weight (kg)	:	142
8.	Impeller		
8.1	Type	:	Closed
8.2	Diameter (mm)	:	89
8.3	Width (mm)	:	31
8.4	No. of vanes	:	4
8.5	Material	:	Brass
9.	Suction Inlet		
9.1	Type	:	Threaded flange
9.2	Diameter (mm)	:	75
9.3	Material	:	Cast Iron
10.	Discharge Outlet		
10.1	Type	:	Threaded flange
10.2	Diameter (mm)	:	75
10.3	Material	:	Cast Iron

Table 1. (Cont'n)

11.	Casing		
11.1	Type	:	Volute
11.2	Material	:	Cast Iron
12.	Stuffing box	:	Packing Seal
13.	Rotation	:	Clockwise when facing the suction inlet
14.	Rated speed (rpm)	:	2200
15.	Rated Maximum Discharge Capacity (L/s)	:	21.7
16.	Rated Maximum Total Head (m)	:	25.0
17.	Primemover (based on Plate Rating)		
17.1	Brand	:	KUBOTA
17.2	Model	:	RK70-S
17.3	Serial No.	:	No Data
17.4	Make	:	P.T. KUBOTA INDONESIA Indonesia
17.5	Type	:	Single cylinder, 4-stroke, water-cooled, horizontal cylinder diesel engine
17.6	Rated Output Power (kW)		
	Maximum	:	5.22 @ 2400 rpm
	Continuous	:	4.48 @ 2200 rpm
17.8	Bore x Stroke (mm)	:	No Data
17.9	Displacement Volume (cc)	:	376
17.10	Cooling System	:	Water-cooled
17.11	Lubricating System	:	Force-feed
17.12	Starting System	:	Rope-recoil
17.13	Air Cleaner	:	Oil Bath type
17.14	Exhaust System	:	Muffler

PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

Pumpset Tested : EAGLE NS80 coupled to KUBOTA RK70-S diesel engine

Date of Test : September 30, 2013

Test Condition

Ambient Temperature (°C) : 30.6

Atmospheric Pressure (mb) : 1013

Pumping Water Temperature (°C) : 27.2

Table 2. Results of Performance Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	FUEL CONSUMP- TION (L/h)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)	WATER POWER (kW)	INPUT POWER (kW)	SYSTEM EFFICIENCY (%)
2173	2475	2.12	24.7	0	0	22.1	0
2120	2417	1.79	22.3	0.45	0.10	18.6	0.53
2072	2297	1.62	20.2	2.89	0.57	16.9	3.39
2055	2334	1.67	18.8	3.51	0.65	17.4	3.37
2037	2301	1.66	17.6	4.77	0.82	17.3	4.76
1991	2260	1.68	15.6	6.53	1.00	17.5	5.70
1959	2220	1.78	13.7	8.11	1.09	18.5	5.88
1922	2188	1.89	11.8	9.41	1.08	19.6	5.52
1880	2143	1.97	8.56	11.0	0.92	20.5	4.49
1818	2075	1.99	5.00	13.8	0.67	20.7	3.27

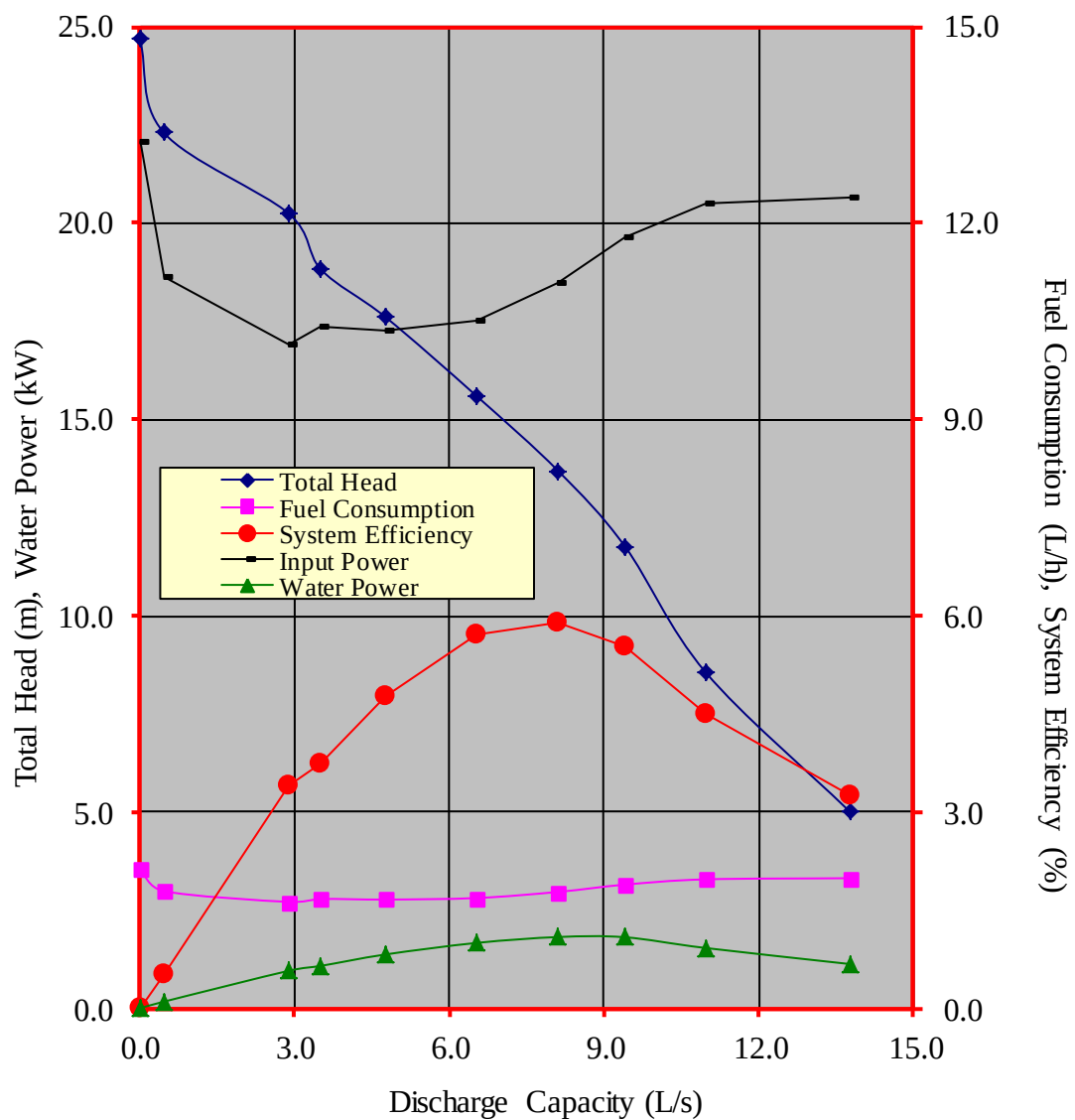


Figure 2. Performance characteristic curves of the EAGLE NS80 coupled to KUBOTA RK70-S diesel engine centrifugal pump

CAVITATION TEST

Results of Cavitation Test are shown in **Table 3**.

Pumpset Tested	:	EAGLE NS80 coupled to KUBOTA RK70-S diesel engine
Date of Test	:	September 30, 2013
Test Condition		
Ambient Temperature (°C)	:	31.6
Atmospheric Pressure (mb)	:	1013
Pumping Water Temperature (°C)	:	27.2

Table 3. Results of Cavitation Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	TOTAL SUCTION LIFT (m)	NET POSITIVE SUCTION HEAD AVAILABLE (m)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)
1853	2110	1.61	8.26	37.0	13.0
1866	2126	3.02	6.85	34.4	12.2
1894	2150	4.42	5.45	32.8	11.6
1928	2183	5.84	4.03	32.1	10.5
2000	2275	6.65	3.22	30.9	7.6
2130	2438	7.47	2.40	28.5	0

DATA COMPUTATION

ANNEX 1. Formula used during calculation and testing

1. Total Head (m), H

$$TDH = H_d - H_s$$

$$\text{Where: } H_d = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2$$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

H_d = Total Discharge Head (m)

H_s = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$ = Pressure gage reading on the discharge pipe at
gage connection converted to meter

$\frac{P_1}{\gamma}$ = Vacuum gage reading on the suction pipe at the
point of gage connection converted to meter

$\frac{V_2^2}{2g}$ = Velocity head on the discharge pipe converted
to meter. Computed by determining the velocity
of liquid on the pipe by formula, $V = Q/A$ where
 V , velocity of liquid inside the pipe in meter per
per second, Q , Discharge rate of the pump in
liters per second, and A , Cross-sectional area
of the pipe in square meter.

$\frac{V_1^2}{2g}$ = Velocity head on the suction pipe converted to
meter

Annex (Cont'n)

Z_2 = The distance from the center of the pressure gage face to the centerline of the pump in meters

Z_1 = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity, Q , in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.
3. Water Power, WP , is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \times Q}{102}$$

4. Input Power, IP , is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determined by the formula:

$$IP = \frac{T \times N}{974}$$

Where:

T = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

N = Input shaft angular speed in RPM

5. Pump Efficiency, η ,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, $NPSHA$. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

P_{atm} = Atmospheric pressure during the test converted to meters

P_{vp} = Vapor pressure of the test liquid at certain temperature converted to meters

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N.B.

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